

B-10-Y

Roll No.....

Total No. of Questions : 16]

[Total No. of Printed Pages : 4

12thARM(SZ)JKUT2024

1110-Y

BOTANY

(Biology)

Time : 1½ Hours]

[Maximum Marks : 35

SECTION-A

1 each

YASIR

1. In RNA, every nucleotide residue has an additional —OH group

presents at :

(A) 1'-position

(B) 2'-position

(C) 3'-position

(D) 5'-position

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Turn Over

2. Bt cotton is not :

- (A) A GM product
- (B) Insect resistant
- (C) A bacterial gene expressing system
- (D) Resistant to all pesticides

3. Which one is Biofertilizer ?

- (A) Mycorrhiza
- (B) Nitrogen fixing bacteria
- (C) Nitrogen fixing cyanobacteria
- (D) All of these

4. Which is not an *in-situ* method of conservation ?

- (A) Biosphere reserves
- (B) Wildlife sanctuary
- (C) National parks
- (D) Cryopreservation

5. An orchid growing as an epiphyte on a mango branch represents an example of :

- (A) Mutualism
- (B) Commensalism
- (C) Protocooperation
- (D) Ammensalism

SECTION-B

2 each

6. Define the terms with one example for each :

- (i) Cleistogamy
- (ii) Epihydrophily

7. What is Double fertilization ?

8. Write a note on various types of processes that occur simultaneously during decomposition of detritus.

9. What is the key difference between Primary and Secondary sewage treatment ?

10. Draw schematic structure of a Transcription unit.

SECTION C

3 each

11. Explain chromosomal theory of inheritance.
12. What is Tissue Culture ? Give its applications.
13. Flowers have evolved various strategies to prevent self pollination.

Write a note on those strategies.

14. What are Hotspots ? Give an example and also write criteria for hotspots.
15. What are properties of genetic material ?

SECTION-D

5

16. Describe the attributes that populations possess but not individuals.

Or

Define Ecological Pyramid. Name the three types of ecological pyramids. What information is conveyed by each pyramid regarding structure, functions and energy in the ecosystem ?